



Firepower Threat Defense High Availability

The following topics describe how to configure Active/Standby failover to accomplish high availability of the Cisco Firepower Threat Defense.

- [About Firepower Threat Defense High Availability, page 1](#)
- [Configure Firepower Threat Defense High Availability, page 12](#)
- [Configure Optional High Availability Parameters, page 13](#)
- [Manage High Availability, page 16](#)
- [Monitoring High Availability, page 20](#)

About Firepower Threat Defense High Availability

Configuring high availability, also called failover, requires two identical Firepower Threat Defense devices connected to each other through a dedicated failover link and, optionally, a state link. Firepower Threat Defense supports Active/Standby failover, where one unit is the active unit and passes traffic. The standby unit does not actively pass traffic, but synchronizes configuration and other state information from the active unit. When a failover occurs, the active unit fails over to the standby unit, which then becomes active.

The health of the active unit (hardware, interfaces, software, and environmental status) is monitored to determine if specific failover conditions are met. If those conditions are met, failover occurs.



Note

High availability is not supported on Firepower Threat Defense Virtual running on Amazon Web Services.

High Availability System Requirements

This section describes the hardware, software, and license requirements for Firepower Threat Defense devices in a High Availability configuration.

Hardware Requirements

The two units in a High Availability configuration must:

- Be the same model.
- Have the same number and types of interfaces.

If you are using units with different flash memory sizes in your High Availability configuration, make sure the unit with the smaller flash memory has enough space to accommodate the software image files and the configuration files. If it does not, configuration synchronization from the unit with the larger flash memory to the unit with the smaller flash memory will fail.

Software Requirements

The two units in a High Availability configuration must:

- Be in the same firewall mode (routed or transparent).
- Have the same major (first number), minor (second number), and maintenance (third number) software version.
- Be in the same domain or group on the Firepower Management Center.
- Have the same NTP configuration. See [Configure NTP Time Synchronization for Threat Defense](#).
- Be fully deployed on the Firepower Management Center with no uncommitted changes.
- Not have DHCP or PPPoE configured in any of their interfaces.

License Requirements

Firepower Threat Defense devices in a high availability configuration must have the same licenses. Before high availability is established, it does not matter which licenses are assigned to the secondary/standby device. During high availability configuration, the Firepower Management Center releases any unnecessary licenses assigned to the standby device and replaces them with identical licenses assigned to the primary/active device. For example, if the active device has a Base license and a Threat license, and the standby device has only a Base license, the Firepower Management Center communicates with the Cisco Smart Software Manager to obtain an available Threat license from your account for the standby device. If your Smart Licenses account does not include enough purchased entitlements, your account becomes Out-of-Compliance until you purchase the correct number of licenses. High availability configurations require two Smart License entitlements; one for each device in the pair.

Failover and Stateful Failover Links

The failover link and the optional stateful failover link are dedicated connections between the two units. The same interface on both devices should be used for failover and stateful failover links.

Failover Link

The two units in a failover pair constantly communicate over a failover link to determine the operating status of each unit.

Failover Link Data

The following information is communicated over the failover link:

- The unit state (active or standby)
- Hello messages (keep-alives)
- Network link status
- MAC address exchange
- Configuration replication and synchronization

Interface for the Failover Link

You can use any unused data interface (physical, redundant, or EtherChannel) as the failover link. The failover link interface is not configured as a normal networking interface; it exists for failover communication only. This interface can only be used for the failover link (and also for the state link). The Firepower Threat Defense device does not support sharing interfaces between user data and the failover link. A separate physical, EtherChannel, or redundant interface must be used for the failover link.

**Note**

When using an EtherChannel or Redundant Interface as the failover or stateful link, you must confirm that the same port channel with the same member interfaces exists on both devices before establishing high availability.

For a redundant interface used as the failover link, see the following benefits for added redundancy:

- When a failover unit boots up, it alternates between the member interfaces to detect an active unit.
- If a failover unit stops receiving keepalive messages from its peer on one of the member interfaces, it switches to the other member interface.

For an EtherChannel used as the failover link, to prevent out-of-order packets, only one interface in the EtherChannel is used. If that interface fails, then the next interface in the EtherChannel is used. You cannot alter the EtherChannel configuration while it is in use as a failover link.

Connecting the Failover Link

Connect the failover link in one of the following two ways:

- Using a switch, with no other device on the same network segment (broadcast domain or VLAN) as the failover interfaces of the Firepower Threat Defense device.
- Using an Ethernet cable to connect the units directly, without the need for an external switch.

If you do not use a switch between the units, if the interface fails, the link is brought down on both peers. This condition may hamper troubleshooting efforts because you cannot easily determine which unit has the failed interface and caused the link to come down.

The Firepower Threat Defense device supports Auto-MDI/MDIX on its copper Ethernet ports, so you can either use a crossover cable or a straight-through cable. If you use a straight-through cable, the interface automatically detects the cable and swaps one of the transmit/receive pairs to MDIX.

Stateful Failover Link

To use Stateful Failover, you must configure a Stateful Failover link (also known as the state link) to pass connection state information.

Shared with the Failover Link

Sharing a failover link is the best way to conserve interfaces. If you experience performance problems on that interface, consider dedicating a separate interface for the state link.

Dedicated Interface for the Stateful Failover Link

You can use a dedicated data interface (physical, redundant, or EtherChannel) for the state link. For an EtherChannel used as the state link, to prevent out-of-order packets, only one interface in the EtherChannel is used. If that interface fails, then the next interface in the EtherChannel is used.

Connect a dedicated state link in one of the following two ways:

- Using a switch, with no other device on the same network segment (broadcast domain or VLAN) as the failover interfaces of the Firepower Threat Defense device.
- Using an Ethernet cable to connect the appliances directly, without the need for an external switch.

If you do not use a switch between the units, if the interface fails, the link is brought down on both peers. This condition may hamper troubleshooting efforts because you cannot easily determine which unit has the failed interface and caused the link to come down.

The Firepower Threat Defense device supports Auto-MDI/MDIX on its copper Ethernet ports, so you can either use a crossover cable or a straight-through cable. If you use a straight-through cable, the interface automatically detects the cable and swaps one of the transmit/receive pairs to MDIX.

For optimum performance when using long distance failover, the latency for the state link should be less than 10 milliseconds and no more than 250 milliseconds. If latency is more than 10 milliseconds, some performance degradation occurs due to retransmission of failover messages.

Avoiding Interrupted Failover and Data Links

We recommend that failover links and data interfaces travel through different paths to decrease the chance that all interfaces fail at the same time. If the failover link is down, the Firepower Threat Defense device can use the data interfaces to determine if a failover is required. Subsequently, the failover operation is suspended until the health of the failover link is restored.

See the following connection scenarios to design a resilient failover network.

Scenario 1—Not Recommended

If a single switch or a set of switches are used to connect both failover and data interfaces between two Firepower Threat Defense devices, then when a switch or inter-switch-link is down, both Firepower Threat Defense devices become active.

Scenario 2—Recommended

We recommend that failover links NOT use the same switch as the data interfaces. Instead, use a different switch or use a direct cable to connect the failover link.

Scenario 3—Recommended

If the Firepower Threat Defense data interfaces are connected to more than one set of switches, then a failover link can be connected to one of the switches, preferably the switch on the secure (inside) side of network, as shown in the following figure.

Scenario 4—Recommended

The most reliable failover configurations use a redundant interface on the failover link.

MAC Addresses and IP Addresses in Failover

When you configure your interfaces, you can specify an active IP address and a standby IP address on the same network. Although recommended, the standby address is not required. Without a standby IP address, the active unit cannot perform network tests to check the standby interface health; it can only track the link state. You also cannot connect to the standby unit on that interface for management purposes.

- 1 When the primary unit fails over, the secondary unit assumes the IP addresses and MAC addresses of the primary unit and begins passing traffic.
- 2 The unit that is now in standby state takes over the standby IP addresses and MAC addresses.

Because network devices see no change in the MAC to IP address pairing, no ARP entries change or time out anywhere on the network.

If the secondary unit boots without detecting the primary unit, the secondary unit becomes the active unit and uses its own MAC addresses, because it does not know the primary unit MAC addresses. However, when the primary unit becomes available, the secondary (active) unit changes the MAC addresses to those of the primary unit, which can cause an interruption in your network traffic. Similarly, if you swap out the primary unit with new hardware, a new MAC address is used.

Virtual MAC addresses guard against this disruption because the active MAC addresses are known to the secondary unit at startup, and remain the same in the case of new primary unit hardware. You can manually configure virtual MAC addresses.

If you do not configure virtual MAC addresses, you might need to clear the ARP tables on connected routers to restore traffic flow. The Firepower Threat Defense device does not send gratuitous ARPs for static NAT addresses when the MAC address changes, so connected routers do not learn of the MAC address change for these addresses.

The IP address and MAC address for the state link do not change at failover; the only exception is if the state link is configured on a regular data interface.

Stateful Failover

During Stateful Failover, the active unit continually passes per-connection state information to the standby unit. After a failover occurs, the same connection information is available at the new active unit. Supported end-user applications are not required to reconnect to keep the same communication session.

Supported Features

The following state information is passed to the standby Firepower Threat Defense device when Stateful Failover is enabled:

- NAT translation table
- TCP connection states
- UDP connection states
- Snort connection states
- Strict TCP enforcement
- The ARP table
- The Layer 2 bridge table (for bridge groups)
- SIP signalling sessions
- Snort Inspection
- Static Routes
- Dynamic Routing Protocols—Stateful Failover participates in dynamic routing protocols, like OSPF and EIGRP, so routes that are learned through dynamic routing protocols on the active unit are maintained in a Routing Information Base (RIB) table on the standby unit. Upon a failover event, packets travel normally with minimal disruption to traffic because the active secondary Firepower Threat Defense device initially has rules that mirror the primary Firepower Threat Defense device. Immediately after failover, the re-convergence timer starts on the newly Active unit. Then the epoch number for the RIB table increments. During re-convergence, OSPF and EIGRP routes become updated with a new epoch number. Once the timer is expired, stale route entries (determined by the epoch number) are removed from the table. The RIB then contains the newest routing protocol forwarding information on the newly Active unit.



Note

Routes are synchronized only for link-up or link-down events on an active unit. If the link goes up or down on the standby unit, dynamic routes sent from the active unit may be lost. This is normal, expected behavior.

- DHCP Server
- ARP Inspection
- AVC—App-id verdicts are replicated, but not detection states. So, proper synchronization occurs as long as App-id verdicts are complete and synchronized before failover occurs.
- URL
- Geolocation
- URL Filtering
- IPS Detection state—Upon failover, once mid-flow pickup occurs, new inspections are completed, but old states are lost.
- File malware blocking—File disposition must become available before failover.

- File type detection—File type must be identified before failover.
- TLS sessions not decrypted
- TLS URL
- User Agent
- ISE Session Directory
- Identity/Captive Portal—Existing user sessions work correctly after failover. For authentications in progress when failover occurs, depending on the race condition between captive portal flows and the failover switch, there are two possible outcomes:
 - The user has not authenticated before failover occurs. In this case, the browser session fails. Upon refreshing, the session goes to the new active unit and you are returned to the captive portal configuration page.
 - The user already authenticated with the original active unit, but failover occurs and the browser session fails. Upon refreshing, you are returned to the captive portal configuration page so you can authenticate a second time with the new active unit.
- Signature Lookup—If failover occurs in the middle of a file transmit, no file event is generated and no detection occurs.
- File Storage—If failover occurs when the file is being stored, it is stored on the original active device. If the original active device went down while the file was being stored, the file does not get stored.
- File Pre-class (Local Analysis)—If failover occurs in the middle of pre-classification, detection fails.
- File Dynamic Analysis - Connectivity to the cloud—If failover occurs, the system may submit the file to the cloud.
- Archive File Support—If fail over occurs in the middle of an analysis, the system loses visibility into the file/archive.
- Custom Blacklisting—If failover occurs, no events are generated.
- IP Reputation
- URL Reputation
- DNS Sinkhole
- Fragment settings

Unsupported Features

The following state information is *not* passed to the standby Firepower Threat Defense device when Stateful Failover is enabled:

- Sessions inside plaintext tunnels
- Inspection after decryption
- TLS Decryption State
- The HTTP connection table

- DHCP client
- DHCP server address leases
- Multicast routing

Transparent Firewall Mode Bridge Group Requirements for Failover

There are special considerations for failover when using bridge groups. When the active unit fails over to the standby unit, the connected switch port running Spanning Tree Protocol (STP) can go into a blocking state for 30 to 50 seconds when it senses the topology change. To avoid traffic loss while the port is in a blocking state, you can configure one of the following workarounds:

- Disable interface monitoring.
- Increase interface holdtime to a high value that will allow STP to converge before the Firepower Threat Defense devices fail over.
- Decrease STP timers to allow STP to converge faster than the interface holdtime.

Failover Health Monitoring

The Firepower Threat Defense device monitors each unit for overall health and for interface health. This section includes information about how the Firepower Threat Defense device performs tests to determine the state of each unit.

Unit Health Monitoring

The Firepower Threat Defense device determines the health of the other unit by monitoring the failover link with hello messages. When a unit does not receive three consecutive hello messages on the failover link, the unit sends LANTEST messages on each data interface, including the failover link, to validate whether or not the peer is responsive. The action that the Firepower Threat Defense device takes depends on the response from the other unit. See the following possible actions:

- If the Firepower Threat Defense device receives a response on the failover link, then it does not fail over.
- If the Firepower Threat Defense device does not receive a response on the failover link, but it does receive a response on a data interface, then the unit does not failover. The failover link is marked as failed. You should restore the failover link as soon as possible because the unit cannot fail over to the standby while the failover link is down.
- If the Firepower Threat Defense device does not receive a response on any interface, then the standby unit switches to active mode and classifies the other unit as failed.

Interface Monitoring

When a unit does not receive hello messages on a monitored interface for 2 polling periods, it runs interface tests. If all interface tests fail for an interface, but this same interface on the other unit continues to successfully pass traffic, then the interface is considered to be failed. If the threshold for failed interfaces is met, then a

failover occurs. If the other unit interface also fails all the network tests, then both interfaces go into the “Unknown” state and do not count towards the failover limit.

An interface becomes operational again if it receives any traffic. A failed device returns to standby mode if the interface failure threshold is no longer met.

If an interface has IPv4 and IPv6 addresses configured on it, the device uses the IPv4 addresses to perform the health monitoring.

If an interface has only IPv6 addresses configured on it, then the device uses IPv6 neighbor discovery instead of ARP to perform the health monitoring tests. For the broadcast ping test, the device uses the IPv6 all nodes address (FE02::1).

Interface Tests

The Firepower Threat Defense device uses the following interface tests:

- 1 Link Up/Down test—A test of the interface status. If the Link Up/Down test indicates that the interface is down, then the device considers it failed. If the status is Up, then the device performs the Network Activity test.
- 2 Network Activity test—A received network activity test. The purpose of this test is to generate network traffic using LANTEST messages to determine which (if either) unit has failed. At the start of the test, each unit clears its received packet count for its interfaces. As soon as a unit receives any packets during the test (up to 5 seconds), then the interface is considered operational. If one unit receives traffic and the other unit does not, then the unit that received no traffic is considered failed. If neither unit received traffic, then the device starts the ARP test.
- 3 ARP test—A reading of the unit ARP cache for the 2 most recently acquired entries. One at a time, the unit sends ARP requests to these machines, attempting to stimulate network traffic. After each request, the unit counts all received traffic for up to 5 seconds. If traffic is received, the interface is considered operational. If no traffic is received, an ARP request is sent to the next machine. If at the end of the list no traffic has been received, the device starts the ping test.
- 4 Broadcast Ping test—A ping test that consists of sending out a broadcast ping request. The unit then counts all received packets for up to 5 seconds. If any packets are received at any time during this interval, the interface is considered operational and testing stops. If no traffic is received, the testing starts over again with the ARP test.

Interface Status

Monitored interfaces can have the following status:

- Unknown—Initial status. This status can also mean the status cannot be determined.
- Normal—The interface is receiving traffic.
- Normal (Waiting)—The interface is up, but has not yet received a hello packet from the corresponding interface on the peer unit.
- Normal (Not-Monitored)—The interface is up, but is not monitored by the failover process.
- Testing—Hello messages are not heard on the interface for five poll times.
- Link Down—The interface or VLAN is administratively down.

- Link Down (Waiting)—The interface or VLAN is administratively down and has not yet received a hello packet from the corresponding interface on the peer unit.
- Link Down (Not-Monitored)—The interface or VLAN is administratively down, but is not monitored by the failover process.
- No Link—The physical link for the interface is down.
- No Link (Waiting)—The physical link for the interface is down and has not yet received a hello packet from the corresponding interface on the peer unit.
- No Link (Not-Monitored)—The physical link for the interface is down, but is not monitored by the failover process.
- Failed—No traffic is received on the interface, yet traffic is heard on the peer interface.

Failover Triggers and Detection Timing

The following table shows the failover triggering events and associated failure detection timing. If failover occurs, you can view the reason for the failover in the Message Center, along with various operations pertaining to the high availability pair.

Table 1: Firepower Threat Defense Failover Times

Failover Triggering Event	Minimum	Default	Maximum
Active unit loses power or stops normal operation.	800 milliseconds	15 seconds	45 seconds
Active unit interface physical link down.	500 milliseconds	5 seconds	15 seconds
Active unit interface up, but connection problem causes interface testing.	5 seconds	25 seconds	75 seconds

About Active/Standby Failover

Active/Standby failover lets you use a standby Firepower Threat Defense device to take over the functionality of a failed unit. When the active unit fails, it changes to the standby state while the standby unit changes to the active state.

Primary/Secondary Roles and Active/Standby Status

When setting up Active/Standby failover, you configure one unit to be primary and the other as secondary. During configuration, the primary unit's policies are synchronized to the secondary unit. At this point, the two units act as a single device for device and policy configuration. However, for events, dashboards, reports and health monitoring, they continue to display as separate devices.

The main differences between the two units in a failover pair are related to which unit is active and which unit is standby, namely which IP addresses to use and which unit actively passes traffic.

However, a few differences exist between the units based on which unit is primary (as specified in the configuration) and which unit is secondary:

- The primary unit always becomes the active unit if both units start up at the same time (and are of equal operational health).
- The primary unit MAC addresses are always coupled with the active IP addresses. The exception to this rule occurs when the secondary unit becomes active and cannot obtain the primary unit MAC addresses over the failover link. In this case, the secondary unit MAC addresses are used.

Active Unit Determination at Startup

The active unit is determined by the following:

- If a unit boots and detects a peer already running as active, it becomes the standby unit.
- If a unit boots and does not detect a peer, it becomes the active unit.
- If both units boot simultaneously, then the primary unit becomes the active unit, and the secondary unit becomes the standby unit.

Failover Events

In Active/Standby failover, failover occurs on a unit basis.

The following table shows the failover action for each failure event. For each failure event, the table shows the failover policy (failover or no failover), the action taken by the active unit, the action taken by the standby unit, and any special notes about the failover condition and actions.

Table 2: Failover Events

Failure Event	Policy	Active Group Action	Standby Group Action	Notes
Active unit failed (power or hardware)	Failover	n/a	Become active Mark active as failed	No hello messages are received on any monitored interface or the failover link.
Formerly active unit recovers	No failover	Become standby	No action	None.
Standby unit failed (power or hardware)	No failover	Mark standby as failed	n/a	When the standby unit is marked as failed, then the active unit does not attempt to fail over, even if the interface failure threshold is surpassed.
Failover link failed during operation	No failover	Mark failover link as failed	Mark failover link as failed	You should restore the failover link as soon as possible because the unit cannot fail over to the standby unit while the failover link is down.

Failure Event	Policy	Active Group Action	Standby Group Action	Notes
Failover link failed at startup	No failover	Mark failover link as failed	Become active	If the failover link is down at startup, both units become active.
State link failed	No failover	No action	No action	State information becomes out of date, and sessions are terminated if a failover occurs.
Interface failure on active unit above threshold	Failover	Mark active as failed	Become active	None.
Interface failure on standby unit above threshold	No failover	No action	Mark standby as failed	When the standby unit is marked as failed, then the active unit does not attempt to fail over even if the interface failure threshold is surpassed.

Configure Firepower Threat Defense High Availability

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

When establishing an Active/Standby High Availability pair, you designate one of the devices as primary and the other as secondary. The system applies a merged configuration to the paired devices. If there is a conflict, the system applies the configuration from the device you designated as primary.

In a multidomain deployment, devices in a high availability pair must belong to the same domain.



Note

The system uses the failover link to sync configuration, while the stateful failover link is used to sync application content between peers. After high availability is established, selected interface links and encryption settings cannot be modified without breaking the high availability pair and reconfiguring it.

Before You Begin

Confirm that both devices:

- Are the same model.
- Have the same number and type of interfaces.
- Are in the same domain and group.

- Have normal health status and are running the same software.
- Are either in routed or transparent mode.
- Have the same NTP configuration. See [Configure NTP Time Synchronization for Threat Defense](#).
- Are fully deployed with no uncommitted changes.
- Do not have DHCP or PPPoE configured in any of their interfaces.

Procedure

-
- Step 1** Choose **Devices > Device Management**.
 - Step 2** From the **Add** drop-down menu, choose **Add High Availability**.
 - Step 3** Enter a display **Name** for the high availability pair.
 - Step 4** Under **Device Type**, choose **Firepower Threat Defense**.
 - Step 5** Choose the **Primary Peer** device for the high availability pair.
 - Step 6** Choose the **Secondary Peer** device for the high availability pair.
 - Step 7** Click **Continue**.
 - Step 8** Under LAN Failover Link, choose an **Interface** with enough bandwidth to reserve for failover communications.
 - Step 9** Type any identifying **Logical Name**.
 - Step 10** Type a **Primary IP** address for the failover link on the active unit. This address should be on an unused subnet.
 - Step 11** Optionally, choose **Use IPv6 Address**.
 - Step 12** Type a **Secondary IP** address for the failover link on the standby unit. This IP address must be in the same subnet as the primary IP address.
 - Step 13** If IPv4 addresses are used, type a **Subnet Mask** that applies to both the primary and secondary IP addresses.
 - Step 14** Optionally, under Stateful Failover Link, choose the same **Interface**, or choose a different interface and enter the high availability configuration information.
 - Step 15** Optionally, choose **Enabled** and choose the **Key Generation** method for IPsec Encryption between the failover links.
 - Step 16** Click **OK**. This process takes a few minutes as the process synchronizes system data.
-

Configure Optional High Availability Parameters

You can view the initial High Availability Configuration on the Firepower Management Center. You cannot edit these settings without breaking the high availability pair and then re-establishing it.

You can edit the Failover Trigger Criteria to improve failover results. Interface Monitoring allows you to determine which interfaces are better suited for failover.

Configure Interface Monitoring

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

By default, monitoring is enabled on all physical interfaces with logical names configured. You might want to exclude interfaces attached to less critical networks from affecting your failover policy.

Procedure

-
- Step 1** Choose **Devices > Device Management**.
- Step 2** Next to the device high-availability pair you want to edit, click the edit icon (✎).
In a multidomain deployment, if you are not in a leaf domain, the system prompts you to switch.
- Step 3** Choose **High Availability**.
- Step 4** Under Monitor Interfaces, click the edit icon (✎).
- Step 5** Choose **Monitor this interface for failures**.
- Step 6** Click **OK**.
-

Edit High Availability Failover Criteria

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

You can customize failover criteria based on your network deployment.

Procedure

-
- Step 1** Choose **Devices > Device Management**.
- Step 2** Next to the device high-availability pair you want to edit, click the edit icon (✎).
In a multidomain deployment, if you are not in a leaf domain, the system prompts you to switch.

- Step 3** Choose **High Availability**.
- Step 4** Next to Failover Trigger Criteria, click the edit icon (✎).
- Step 5** Under Interface Failure Threshold, choose the number or percentage of interfaces that must fail before the device fails over.
- Step 6** Under Hello packet Intervals, choose how often hello packets are sent over the failover link.
- Step 7** Click **OK**.

Configure Virtual MAC addresses

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

You can configure active and standby MAC addresses for failover in two places on the Firepower Management Center:

- The Advanced tab of the Edit Interface page during interface configuration; see [Configure the MAC Address](#).
- The Add Interface MAC Address page accessed from the High Availability page; see

If active and standby MAC addresses are configured in both locations, the addresses defined during interface configuration takes preference for failover.

You can minimize loss of traffic during failover by designating active and standby mac addresses to the physical interface. This feature offers redundancy against IP address mapping for failover.

Procedure

- Step 1** Choose **Devices > Device Management**.
- Step 2** Next to the device high-availability pair you want to edit, click the edit icon (✎).
In a multidomain deployment, if you are not in a leaf domain, the system prompts you to switch.
- Step 3** Choose **High Availability**.
- Step 4** Choose the add icon (+) next to Interface Mac Addresses.
- Step 5** Choose a **Physical Interface**.
- Step 6** Type an **Active Interface Mac Address**.
- Step 7** Type a **Standby Interface Mac Address**.
- Step 8** Click **OK**.

Manage High Availability

This section describes how to manage High Availability units after you enable High Availability, including how to change the High Availability setup and how to force failover from one unit to another.

Switch the Active Peer in a Firepower Threat Defense High Availability Pair

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

After you establish a Firepower Threat Defense high availability pair, you can manually switch the active and standby units, effectively forcing failover for reasons such as persistent fault or health events on the current active unit. Both units should be fully deployed before you complete this procedure.

Procedure

-
- Step 1** Choose **Devices > Device Management**.
- Step 2** Next to the high availability pair where you want to change the active peer, click the Switch Active Peer icon ()
- Step 3** You can:
- Click **Yes** to immediately make the backup device the active device in the high availability pair.
 - Click **No** to cancel and return to the Device Management page.
-

Suspend and Resume High Availability

Using CLI, you can suspend a unit in a high availability pair. This feature becomes particularly useful when:

- Both units are in an active-active situation and you want to suspend one of them.
- You want to troubleshoot an active or standby unit and do not want the units to fail over during that time.

Use the following CLI command on a particular unit of the high availability pair to suspend failover:

```
configure failover suspend
```


This command checks the high availability configuration and then suspends it. If the device is in an active state, failover occurs if it is connected to a standby unit. If the unit is an active node without a peer node, it goes into a suspended state, where it no longer processes traffic. If the unit is a standby unit, it goes into a suspended state.

To resume failover, use the following command:

```
configure failover resume
```

**Note**

Failover suspension is not a permanent state. If you reload the unit, it resumes automatically and returns to a standalone unit.

Replace a Unit

If you need to replace a failed unit in a Firepower Threat Defense high availability pair, you must choose the Force Break option to separate the pair. After you replace or repair the unit, you must then register the device on the Firepower Management Center and re-establish high availability. The process varies depending on whether the device is primary or secondary.

See the following topics for more information:

- [Replace a Primary Unit, on page 17](#)
- [Replace a Secondary Unit, on page 18](#)

Replace a Primary Unit

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

Follow the steps below to replace a failed primary unit in a Firepower Threat Defense high availability pair. Failing to follow these steps can overwrite the existing high availability configuration.

Procedure

- Step 1** Choose **Force Break** to separate the high availability pair; see [Separate Units in a High Availability Pair, on page 19](#).
The secondary unit becomes active and retains the failover configuration.

- Step 2** Unregister the failed primary Firepower Threat Defense device from the Firepower Management Center; see [Deleting Devices from the Firepower Management Center](#).
- Step 3** Register the replacement Firepower Threat Defense to the Firepower Management Center; see [Adding Devices to the Firepower Management Center](#).
- Step 4** Configure high availability, using the existing secondary/active unit as the primary device and the replacement device as the secondary/standby device during registration; see [Configure Firepower Threat Defense High Availability](#), on page 12.

What to Do Next

Replace a Secondary Unit

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

Follow the steps below to replace a failed secondary unit in a Firepower Threat Defense high availability pair.

Procedure

- Step 1** Choose **Force Break** to separate the high availability pair; see [Separate Units in a High Availability Pair](#), on page 19.
The primary unit becomes active and retains the failover configuration.
- Step 2** Unregister the secondary Firepower Threat Defense device from the Firepower Management Center; see [Deleting Devices from the Firepower Management Center](#).
- Step 3** Register the replacement Firepower Threat Defense to the Firepower Management Center; see [Adding Devices to the Firepower Management Center](#).
- Step 4** Configure high availability, using the existing primary/active unit as the primary device and the replacement device as the secondary/standby device during registration; see [Configure Firepower Threat Defense High Availability](#), on page 12.

Separate Units in a High Availability Pair

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

When you break a high availability pair, the active device retains full deployment functionality. The standby device loses its failover and interface configurations, and becomes a standalone device.

**Note**

If you cannot reach the high availability pair using the Firepower Management Center, use the CLI command `configure failover disable` to remove the failover configuration from both devices.

Procedure

- Step 1** Choose **Devices > Device Management**.
- Step 2** Next to the high-availability pair you want to break, click the Break HA icon (.
- Step 3** Optionally, check the check box to force break, if the standby peer does not respond.
- Step 4** Click **Yes**. The device high-availability pair is separated.

Unregister a High Availability Pair

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

You can delete the pair from the Firepower Management Center and disable High Availability on each unit using the CLI.

Before You Begin

This procedure requires CLI access.

Procedure

Step 1 Choose **Devices > Device Management**.

Step 2 Next to the high-availability pair you want to unregister, click the Delete icon ()

Step 3 Click **Yes**. The device high availability pair is deleted.

Step 4 On each unit, access the Firepower Threat Defense CLI, and enter the following command:
configure high-availability disable

If you do not enter this command, you cannot re-register the units and form a new HA pair.

Note Enter this command *before* you change the firewall mode; if you change the mode, the unit will not later let you enter the **configure high-availability disable** command, and the Firepower Management Center cannot re-form the HA pair without this command.

Monitoring High Availability

This section lets you monitor the High Availability status.

View Failover History

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

You can view the failover history of both high availability devices in a single view. The history displays in chronological order and includes the reason for any failover.

Procedure

Step 1 Choose **Devices > Device Management**.

Step 2 Next to the device high-availability pair you want to edit, click the edit icon ().
In a multidomain deployment, if you are not in a leaf domain, the system prompts you to switch.

Step 3 Choose **Summary**.

Step 4 Under General, click the view icon ().

View Stateful Failover Statistics

Smart License	Classic License	Supported Devices	Supported Domains	Access
Any	N/A	Firepower Threat Defense Firepower Threat Defense Virtual	Any	Admin/Network Admin

You can view the stateful failover link statistics of both the primary and secondary devices in the high availability pair.

Procedure

-
- Step 1** Choose **Devices > Device Management**.
 - Step 2** Next to the device high-availability pair you want to edit, click the edit icon (✎).
In a multidomain deployment, if you are not in a leaf domain, the system prompts you to switch.
 - Step 3** Choose **High Availability**.
 - Step 4** Under Stateful Failover Link, click the view icon (🔍).
 - Step 5** Choose a device to view statistics.
-

